

Integration with GCP(google cloud) lakehouse runtime catalog with snowflake

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Introduction

GCP(Google cloud) biglake catalog lakehouse runtime catalog , [snowflake](#) .
 , , , , . 2026 7 .

Iceberg lakehouse

Apache Iceberg Apache 2.0 . Iceberg ,
 RDBMS, , DB .
 , iceberg iceberg ,
 . , .
 AWS , S3, Redshift, glue catalog . GCP Google Cloud Stor-
 age(GCS), BigQuery, lakehouse runtime catalog(biglake catalog) . -
 managed catalog , , polaris catalog managed catalog .

GCP

GCP 2 iceberg table . bigquery iceberg table , lakehouse runtime catalog iceberg
 table . , 4 :

- bigquery (bigquery native table)
- bigquery iceberg table(bigquery managed iceberg table)
- lakehouse runtime catalog(biglake) iceberg table
-

bigquery managed iceberg table CREATE iceberg :

```
CREATE TABLE [PROJECT_ID.]DATASET_ID.TABLE_NAME (  
  COLUMN DATA_TYPE[, ...]  
)  
CLUSTER BY CLUSTER_COLUMN_LIST  
WITH CONNECTION {CONNECTION_NAME | DEFAULT}  
OPTIONS (  
  file_format = 'PARQUET',  
  table_format = 'ICEBERG',  
  storage_uri = '<STORAGE_URI>');
```

, lakehouse runtime catalog iceberg table lakehouse runtime catalog , spark . bigquery :

```
CREATE EXTERNAL TABLE `<table name>`  
  WITH CONNECTION `<connection name>`  
  OPTIONS (  
    format = 'ICEBERG',  
    uris = ['<STORAGE_URI>']  
  );
```

1

GCS . , bigquery managed iceberg table bigquery , lakehouse runtime catalog iceberg table .

snowflake lakehouse runtime catalog . , bigquery lakehouse runtime catalog . AWS , glue catalog S3 athena , glue spark .

snowflake

snowflake bigquery , , external volume GCS , bigquery managed iceberg table snowflake metadata . bigquery iceberg table v0.metadata.json metadata.json , iceberg metadata , snowflake .

```
EXPORT TABLE METADATA FROM <table name>
```

, GCS metadata v1776853980.metadata.json metadata avro files . metadata iceberg , snowflake external iceberg table . export external iceberg table . version-hint.text metadata (1776853980) , .

bigquery export metadata metadata , snowflake meta-
data . snowflake , metadata export , metadata s-
nowflake . , GCP snowflake ,
GCP snowflake .
, GA [snowflake lakehouse runtime catalog](#) , lakehouse runtime catalog iceberg
table , :

```
CREATE OR REPLACE CATALOG INTEGRATION biglake_catalog_int
  CATALOG_SOURCE = ICEBERG_REST
  TABLE_FORMAT = ICEBERG
  REST_CONFIG = (
    CATALOG_URI = 'https://biglake.googleapis.com/iceberg/v1/restcatalog'
    CATALOG_NAME = '<gcs_base_location>'
    ADDITIONAL_HEADERS = (
      "x-goog-user-project" = '<gcp_project_id>'
    )
  )
  REST_AUTHENTICATION = (
    TYPE = OAUTH
    OAUTH_GRANT_TYPE = TOKEN_EXCHANGE
    OAUTH_TOKEN_URI = 'https://sts.googleapis.com/v1/token'
    OAUTH_AUDIENCE = '<gcp_oidc_audience_url>'
    OAUTH_ALLOWED_SCOPES = ('https://www.googleapis.com/auth/bigquery')
  )
  ENABLED = TRUE;
```

2

bigquery , 1 bigquery lakehouse runtime catalog .
, bigquery , , bigquery native
table iceberg table .

CATALOG_NAME metadata , , .

proxy server bearer token

proxy server , 2 snowflake lakehouse runtime catalog . 2 :

1. GCP bearer token 1 ,
2. bigquery managed iceberg table lakehouse runtime catalog , snowflake

snowflake catalog integration , Apache Iceberg™ REST . OAuth,
 , SigV4 , GA OAuth ¹, . GCP 1 ,
 . proxy cloud run , snowflake , :

```
import os

import google.auth
import google.auth.transport.requests
import requests
from flask import Flask, Response, abort, request

app = Flask(__name__)
PROJECT_ID = os.environ["GCP_PROJECT_ID"]
PROXY_TOKEN = os.environ["PROXY_BEARER_TOKEN"]
BIGLAKE_BASE = "https://biglake.googleapis.com/iceberg/v1/restcatalog"

def get_gcp_token():
    creds, _ = google.auth.default()
    creds.refresh(google.auth.transport.requests.Request())
    return creds.token

@app.route("/", methods=["GET"])
@app.route("/<path:path>", methods=["GET", "POST", "DELETE", "HEAD"])
def proxy(path=""):
    if request.headers.get("Authorization") != f"Bearer {PROXY_TOKEN}":
        abort(401)
    url = f"{BIGLAKE_BASE}/{path}" if path else BIGLAKE_BASE
    headers = {
        "Authorization": f"Bearer {get_gcp_token()}",
        "x-goog-user-project": PROJECT_ID,
    }
    if request.content_type:
        headers["Content-Type"] = request.content_type
    params = dict(request.args)
    if "warehouse" in params:
        params["warehouse"] = f"bq://projects/{PROJECT_ID}"
    resp = requests.request(
        method=request.method,
        url=url,
        headers=headers,
        params=params,
        data=request.get_data(),
    )
```

¹ . OAuth .

```
)
return Response(resp.content, status=resp.status_code, content_type=resp.headers.get("Co
```

proxy bearer token , GCP () lakehouse runtime catalog .
 PROXY_BEARER_TOKEN GCP , cloud run endpoint (IAM) .
 snowflake catalog integration :

```
CREATE CATALOG INTEGRATION biglake_catalog_int_external
  CATALOG_SOURCE = ICEBERG_REST
  TABLE_FORMAT = ICEBERG
  REST_CONFIG = (
    CATALOG_URI = '<cloud run endpoint of above>'
    CATALOG_NAME = 'gs://<catalog location>'
    ACCESS_DELEGATION_MODE = EXTERNAL_VOLUME_CREDENTIALS
  )
  REST_AUTHENTICATION = (
    TYPE = BEARER
    BEARER_TOKEN = '<self defined bearer token for proxy container>'
  )
  ENABLED = TRUE;
```

snowflake -> proxy container -> lakehouse runtime catalog . snowflake CATALOG_URI ,
 lakehouse runtime catalog REST endpoint , cloud run proxy container endpoint .
 <self defined bearer token for proxy container> . cloud run proxy server ,
 , ADC bearer token , lakehouse runtime catalog REST
 endpoint , , REST endpoint snowflake (x-goog-user-
 project) , 1 snowflake lakehouse runtime catalog .
 , CATALOG_NAME gs://... proxy bq://... . lakehouse runtime cata-
 log , lakehouse runtime catalog bigquery managed iceberg table . catalog
 integration bq://... , snowflake gs://... . proxy container warehouse-
 bq://... , snowflake lakehouse runtime catalog , bigquery managed
 iceberg table .

Summary

- Google cloud biglake catalog lakehouse runtime catalog , snowflake catalog integra-
 tion .
- GCP 4 (bigquery native table, bigquery managed iceberg table, lakehouse runtime
 catalog iceberg table,) , iceberg table (1).
- snowflake 2 . bigquery managed iceberg table EXPORT TABLE METADATA ,
 snowflake external iceberg table , lakehouse runtime catalog catalog integra-
 tion(GA) .

- catalog integration lakehouse runtime catalog , bigquery native/managed table (2).
- cloud run proxy server , (1) GCP 1 bearer token token , (2) catalog federation CATALOG_NAME gs:// bq:// , bigquery managed iceberg table lakehouse runtime catalog snowflake .
- , bigquery snowflake . proxy GCP ,